proteins to be synthesized which makes an organism unique.

Marshall Nirenberg, **Severo Ochoa** (enzyme polynucleotide phosphorylase called Ochoa's enzyme), **Hargobind Khorana**, **Francis Crick** and many others have contributed significantly to decipher the genetic code. The order in which bases are arranged in mRNA decides the order in which amino acids are arranged in proteins. Finally a checker board for genetic code was prepared (**table 5.1**).

The salient features of genetic code are as follows:

- The genetic codon is a **triplet code** and 61 codons code for amino acids and 3 codons do not code for any amino acid and function as **stop codon** (Termination).
- The genetic code is universal. It means that all known living systems use nucleic acids and the same three base codons (triplet codon) direct the synthesis of protein from amino acids. For example, the mRNA (UUU) codon codes for phenylalanine in all cells of all organisms. Some exceptions are reported in prokaryotic, mitochondrial and chloroplast genomes. However similarities are more common than differences.
- A non-overlapping codon means that the same letter is not used for two different codons. For instance, the nucleotide sequence GUU GUC represents only two codons.
- It is comma less, which means that the message would be read directly from one end to the other i.e., no punctuation are needed between two codes.
- A degenerate code means that more than one triplet codon could code for a specific amino acid. For example, codons GUU, GUC, GUA and GUG code for valine.
- Non-ambiguous code means that one codon will code for one amino acid.
- The code is always read in a fixed direction i.e. from 5'→3' direction called polarity.
- AUG has dual functions. It acts as a initiator codon and also codes for the amino acid methionine.

Table 5.1 Genetic code dictionary

Second Nucleotide in Codon

First nucleotide in codon (5' end)	U	C	A	G	Third nucleotide in codon (3' end)
U	UUU Phe F Phenylalanine	UCU Ser S Serine	UAU Tyr Y Tyrosine	UGU Cys C Cysteine	U
	UUC Phe F Phenylalanine	UCC Ser S Serine	UAC Tyr Y Tyrosine	UGC Cys C Cysteine	C
	UUA Leu L Leucine	UCA Ser S Serine	UAA Termination	UGA Termination	A
	UUG Leu L Leucine	UCG Ser S Serine	UAG Termination	UGG Trp W Tryptophan	G
C	CUU Leu L Leucine	CCU Pro P Proline	CAU His H Histidine	CGU Arg R Arginine	U
	CUC Leu L Leucine	CCC Pro P Proline	CAC His H Histidine	CGC Arg R Arginine	C
	CUA Leu L Leucine	CCA Pro P Proline	CAA Gln Q Glutamine	CGA Arg R Arginine	A
	CUG Leu L Leucine	CCG Pro P Proline	CAG Gln Q Glutamine	CGG Arg R Arginine	G
A	AUU Ile I Isoleucine	ACU Thr T Threonine	AAU Asn N Asparagine	AGU Ser S Serine	U
	AUC Ile I Isoleucine	ACC Thr T Threonine	AAC Asn N Asparagine	AGC Ser S Serine	C
	AUA Ile I Isoleucine	ACA Thr T Threonine	AAA Lys K Lysine	AGA Arg R Arginine	A
	AUG Met M Methionine	ACG Thr T Threonine	AAG Lys K Lysine	AGG Arg R Arginine	G
G	GUU Val V Valine	GCU Ala A Alanine	GAU Asp D Aspartic acid	GGU Gly G Glycine	U
	GUC Val V Valine	GCC Ala A Alanine	GAC Asp D Aspartic acid	GGC Gly G Glycine	C
	GUA Val V Valine	GCA Ala A Alanine	GAA Glu E Glutamic acid	GGA Gly G Glycine	A
	GUG Val V Valine	GCG Ala A Alanine	GAG Glu E Glutamic acid	GGG Gly G Glycine	G

Codon

Three-letter and single-letter abbreviations

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